

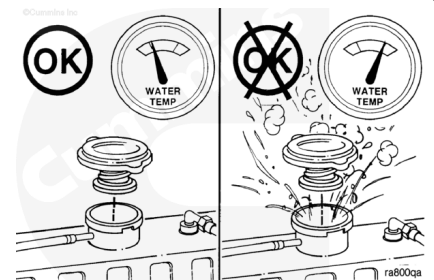
Trainings ▾

General Information

Overflow Method

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Allow the engine to cool, and remove the cooling system pressure cap.

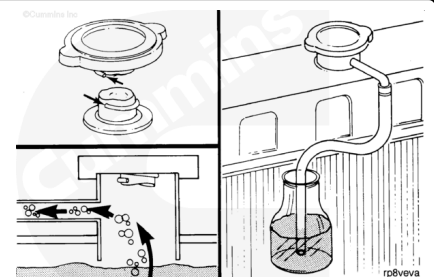
Install a pressure cap that has the spring and the pressure relief valve removed to allow free flow from the overflow tube.

Attach a rubber hose to the radiator overflow connection.

Put the free end of the hose below the water level in a container of water.

Note : The pressure cap must be tightly sealed in the top of the radiator fill neck.

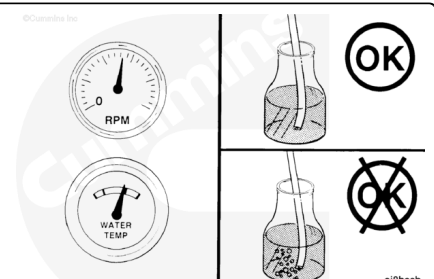
Note : A gasket may need to be installed in order to properly seal this interface. If the pressure cap is not properly sealed then any air bubbles that may be present will not show up during the test.



Operate the engine at rated rpm until it reaches a temperature of 82°C [180°F].

Check for air bubbles in the water container. A combustion gas leak is present if a continuous flow of air bubbles is present or air is entering the coolant system.

Note : The engine coolant temperature must be stable to perform this test. An increasing coolant temperature will give a false indication of air because of the expansion of the coolant in the system.



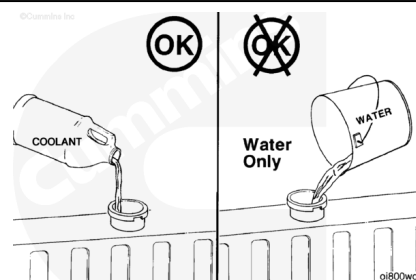
A continuous flow of air bubbles may indicate one of the following:

- A cylinder head combustion gas leak
- Incorrect cylinder liner protrusion
- Cracked or broken cylinder liners
- Damaged aftercooler.

Combustion Gas Leak

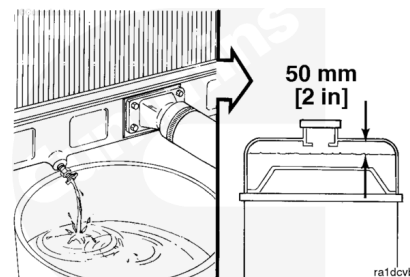
Use combustion gas tester, Part Number 3822985, to test for combustion gases in the cooling system.

Note : It is recommended that the cooling system contain a premixture of 50-percent antifreeze and 50-percent water during the combustion gas leak test. The use of water only can result in a color change in the test fluid from blue to turquoise or light green during the test. This is not an indication of a combustion gas leak.



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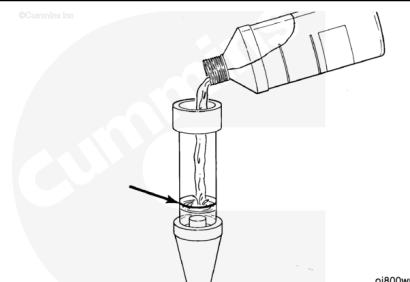
⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

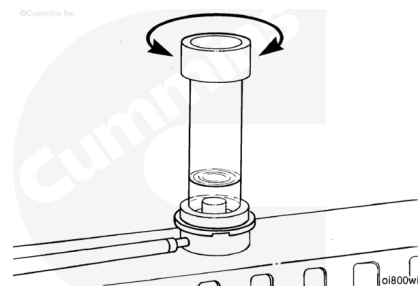
Drain the coolant level down approximately 50 mm [2 in] below the radiator cap seal ledge in the radiator fill neck. Refer to Procedure 008-018 in Section 8.

(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)

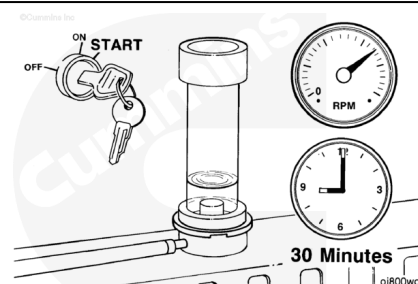
Pour the test fluid into the combustion gas test instrument until it is up to the yellow fill line on the instrument.



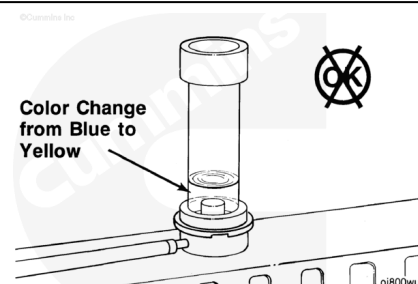
Insert the rubber tip of the combustion gas leak test instrument in the radiator fill neck. Hold the instrument down firmly and turn back and forth to make sure that an airtight seal is formed between the tester and radiator fill neck.



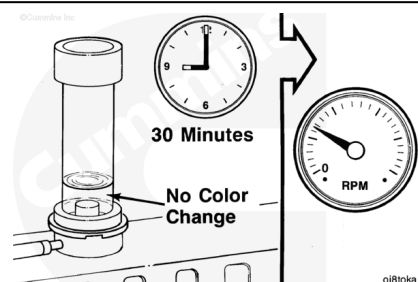
Start the engine and operate at high idle for approximately 30 minutes. Monitor the engine temperature and color of the test fluid during engine operation. Do **not** allow the engine temperature to exceed 100°C [212°F] during the test.



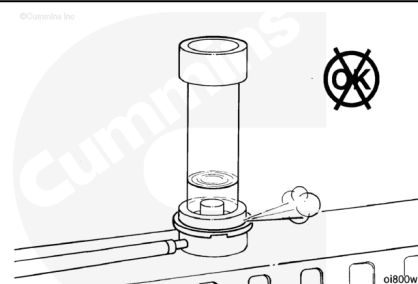
If the color of the test fluid changes from blue to yellow or green anytime during the test, combustion gases are leaking into the cooling system. Discontinue the test if the color of test fluid changes from blue to yellow.



If the color of the test fluid does **not** change from blue to yellow or green during the 30-minute test period, return the engine to low idle.



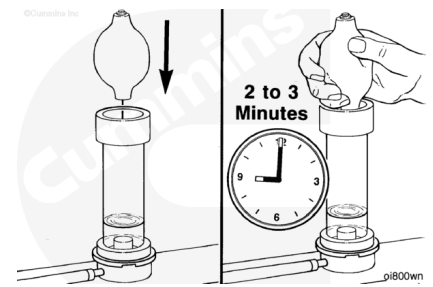
Check the test instrument to make sure it is firmly sealed in the radiator fill neck.



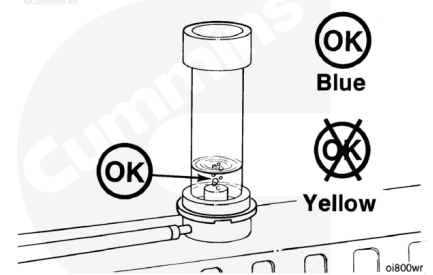
Insert the tip of the rubber ball into the hole in the top of the test instrument. Squeeze the rubber ball 2 to 3 minutes to draw air from the radiator through the test fluid.

If the color of the test fluid remains blue, combustion gases are **not** entering the cooling system.

If the color of the test fluid changes from blue to yellow or green, combustion gases are entering the cooling system and further investigation is required to determine the source of the combustion leak.



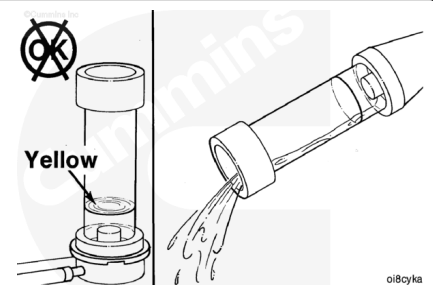
Note : As the cooling system warms up to operating temperature, air will be expelled through the combustion gas tester in the form of bubbles in the test fluid. This is due to normal expansion of the coolant. Do **not** mistake the presence of air bubbles in the tester for combustion gases or air leaking into the cooling system. A change in the color of the test fluid from blue to yellow or green is the **only** indication of combustion gas in the cooling system.



A positive result from the combustion leak tester indicates one or more of the following:

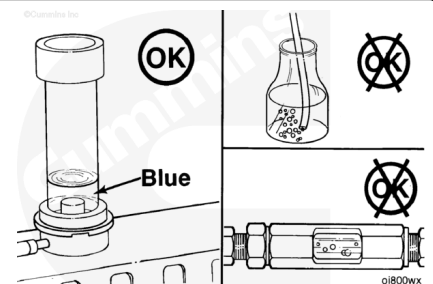
- Cracked cylinder head
- Cylinder head or gasket leakage
- Air compressor head or gasket leak.

Note : Discard the test fluid if it has indicated positive.

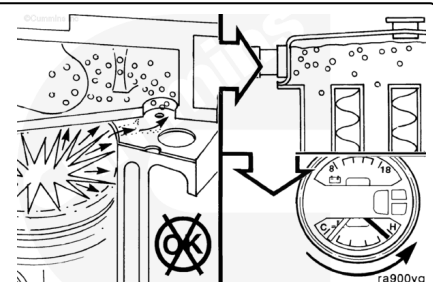


A negative result from the combustion leak tester, coupled with a continuous flow of air bubbles from the previous test, indicates the following:

- Air entrained because of a faulty thermostat jiggle pin, incorrect fill, or faulty coolant system plumbing.



- Air in the coolant can result in loss from the overflow when the aerated coolant is hot. The heated air expands, increasing the pressure in the system and causing the cap to open.
- Similarly, coolant can be displaced through the overflow if the head gasket leaks compression gases into the cooling system.



Note : If the cooling system is contaminated with oil, it **must** be cleaned and flushed. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).

- The operating pressure of the cooling system and the lubricating system can result in the mixing of the fluids if there is a leak between the systems: head gasket and oil cooler.
- Transmission fluid **can** also leak into the cooling system through transmission oil coolers.

